Ex 15

If $f : \mathbb{R}^d \to \mathbb{R}$, $f \in C^1(\mathbb{R}^d)$ and f is strong convex then

(ii) $2D_f(x,y) \leq \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\| \quad \forall x, y \in \mathbb{R}^d$

(iii) $\langle \nabla f(x) - \nabla f(y); x - y \rangle \leq \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\| \quad \forall x, y \in \mathbb{R}^d$

(iii) Let's assume that for a moment (ii) is true then we have

$$\langle \nabla f(x) - \nabla f(y); x - y \rangle = \underline{f(x) - f(x)} + f(y) - \underline{f(y)} + \langle \nabla f(x); y - x \rangle +$$

$$+ \underline{\langle x - y; \nabla f(y) \rangle} +$$

Regrouping them

$$= \Big(f(x) - f(y) - \langle \nabla f(y); x - y \rangle\Big) + \Big(f(y) - f(x) - \langle \nabla f(x); y - x \rangle\Big) =$$

$$= D_f(x,y) + D_f(y,x) \leq \frac{1}{2\mu} \|\nabla f(y) - \nabla f(x)\| + \frac{1}{2\mu} \|\nabla f(x) - \nabla f(y)\|$$

$$= \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\|.$$

(ii) $2D_f(x,y) \leq \frac{1}{\mu} \|\nabla f(x) - \nabla f(y)\| \quad \forall x, y \in \mathbb{R}^d$?

First of all let define $a := \nabla f(x) - \nabla f(y) \quad b = x - y$

then $0 \leq \left\| \frac{a}{\sqrt{\mu}} - \sqrt{\mu} b \right\|^2 = \frac{1}{\mu}\|a\|^2 - 2\langle a; b\rangle + \mu\|x-y\|^2$

~~which means that $\|x-y\|^2 \geq \frac{2}{\mu}\langle \nabla f(x) - \nabla f(y), x-y\rangle - \frac{1}{\mu^2}\|\nabla$~~

which means $\frac{1}{\mu}\|\nabla f(x) - \nabla f(y)\| \geq 2\langle \nabla f(x) - \nabla f(y); x-y\rangle - \mu\|x-y\|^2$ (1)

On the other hand

$f(x) - f(y) - \langle \nabla f(y); x-y\rangle \geq \frac{\mu}{2}\|x-y\|^2$ which is the sum

$f(y) - f(x) \leq \langle \nabla f(y); y-x\rangle - \frac{\mu}{2}\|x-y\|^2$ also

$f(y) - f(x) - \langle \nabla f(x); y-x\rangle \leq \langle \nabla f(y) - \nabla f(x); y-x\rangle - \frac{\mu}{2}\|x-y\|^2$ (by 2)

$D_f(y,x) = f(y) - f(x) - \langle \nabla f(x); y-x\rangle \leq \frac{1}{2\mu}\|\nabla f(x) - \nabla f(y)\|^2$ => a result

$2D_f(y,x) \leq \frac{1}{\mu}\|\nabla f(x) - \nabla f(y)\| \quad \forall x, y \in \mathbb{R}^d$